Research Highlights

- Four packet-level attacker actions are modeled as Lie-cone generators on flows.
- Trajectories in the cone provably preserve rate, length, header and MTU invariants.
- On full IDS2018/2017/UNSW-NB15, FFE evades 99.5/99.2/57.6% of detected attacks.
- FFE beats seven baselines; surrogate CAPGD needs fewer queries but succeeds less.
- Adversarial training helps at its training budget but much less at larger budgets.